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CONSENSUS DOCUMENT

Expert consensus on interstitial lung disease associated with systemic autoimmune diseases. Executive summary

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KEYWORDS
Interstitial lung diseases;

Abstract Interstitial lung disease (ILD) is a frequent and potentially fatal manifestation of systemic autoimmune diseases (SAD). Early screening, diagnosis, and treatment with immunomodulators, and in some cases, antifibrotics, as well as appropriate follow-up, are essential to improving patients' quality of life and survival. The complexity of these diseases

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Systemic sclerosis;
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Inflammatory myopathies;
Anti-Neutrophil cytoplasmic antibody-associated vasculitis;
Consensus

and the limited robust evidence to guide clinical decision-making lead to significant variability in management guidelines. In this article, a group of experts in SAD has developed recommendations for the screening, diagnosis, treatment, and follow-up of ILD, considering the latest evidence, their clinical experience, and the results of a survey conducted among specialists. The diseases included are systemic sclerosis, Sjögren's disease, idiopathic inflammatory myopathy, ANCA-associated vasculitis, sarcoidosis and systemic lupus erythematosus.

PALABRAS CLAVE
Enfermedades pulmonares intersticiales;
Esclerosis sistémica;
Síndrome de sjögren;
Miopatías inflamatorias;
Vasculitis asociada a anticuerpos anti-citoplasma de neutrófilos;
Consenso

Consenso de expertos sobre la enfermedad pulmonar intersticial asociada a enfermedades autoinmunes sistémicas. Resumen ejecutivo

Resumen La enfermedad pulmonar intersticial (EPI) es una manifestación frecuente y potencialmente mortal de las enfermedades autoinmunes sistémicas (EAS). El cribado, diagnóstico y tratamiento precoz con inmunomoduladores y, en algunos casos, antifibróticos, así como un seguimiento adecuado, son fundamentales para mejorar la calidad de vida y la supervivencia de los pacientes. La complejidad de estas enfermedades, junto con la escasa evidencia sólida que guíe la toma de decisiones clínicas, determina una enorme variabilidad en las directrices de manejo. En este artículo, un grupo de expertos en EAS ha elaborado recomendaciones sobre el cribado, diagnóstico, tratamiento y seguimiento de la EPI teniendo en consideración la evidencia más reciente, su experiencia clínica y los resultados de una encuesta realizada a especialistas. Se han incluido esclerosis sistémica, enfermedad de Sjögren, miopatías inflamatorias idiopáticas, vasculitis asociadas a ANCA, sarcoidosis y lupus eritematoso sistémico.

Introduction

Early diagnosis of ILD associated with SAD (ILD-SAD) allows for slowing the progression of pulmonary involvement by initiating immunomodulatory therapy associated, in some cases, with antifibrotic therapy, which has been shown to improve prognosis and patient survival. Therefore, screening, early diagnosis and treatment, and adequate follow-up are essential for improving quality of life and survival. However, clinical practice guidelines and expert recommendations on the management of ILD-SAD vary greatly.[1,2]

In this article, a group of experts has drafted recommendations on the screening, diagnosis, treatment, and follow-up of ILD-SAD, including systemic sclerosis (SS), Sjögren's disease (SjD), idiopathic inflammatory myopathies (IIM), ANCA-associated vasculitis (AAV), and systemic lupus erythematosus (SLE). The recommendations are based on the latest evidence, their clinical experience, and the results of a survey of specialists. Given its importance in internal medicine, sarcoidosis has also been included.

The full consensus document is available in the supplementary material.

Methods

A scientific committee comprising 12 internal medicine specialists with more than ten years of experience in the management of SAD was formed. After a narrative review of the literature and a qualitative assessment of the current evidence to detect the most controversial aspects, the scientific committee created a questionnaire with closed-ended statements in order to reduce the risks in interpreting the answers. These questions reflect the most debated issues on the screening, diagnosis, treatment, and follow-up of ILD-SAD, including SS, SjS, IIM, AAV, and SLE in addition to sarcoidosis.

The literature search was conducted on July 1, 2023 on PubMed and included articles in English and Spanish from the previous five years with the following key terms: *interstitial lung diseases; connective tissue disease-associated interstitial lung disease; autoimmune-associated interstitial lung disease; systemic sclerosis-associated interstitial lung disease; primary Sjögren's disease-associated interstitial lung disease; anti-neutrophil cytoplasmic antibody-associated vasculitis; myositis; polymyositis; dermatomyositis; anti-synthetase syndrome; myositis syndromes; systemic lupus erythematosus-associated interstitial lung disease; sarcoidosis; drug therapy; diagnosis; books and documents, guideline, meta-analysis, review, systematic review.* In addition, a search for guidelines and consensus documents on ILD was performed on the websites of the main scientific societies related to the management of ILD or SAD (see details of this search in the full document in the supplementary material).

In a second phase of the project, the members of the panel of experts were chosen to answer the questions, via the internet, that were drafted by the scientific committee. The panelists (list of panelists in supplementary material) were specialists in internal medicine or pulmonology. They were selected by the committee members according to the criteria of experience and degree of knowledge or connection to SAD and ILD in their professional field. All were members of working groups in pulmonology or internal medicine scientific societies related to SAD or ILD.

A question was considered agreed on if it received the votes of more than 75% of the panelists, with the objective of reflecting significant, realistic agreement and so as not to impede progress on the consensus due to minor disagreements. Taking into account the scientific evidence, issues that were agreed on, and their own clinical experience, the scientific committee wrote recommendations for the management of ILD-SAD.

The ATS/ERS/JRS/ALAT clinical practice guidelines,[2] the definition of progressive fibrosing ILD from the INBUILD study,[5] and the definition by Goh et al.[6] regarding the progression of ILD in SS were used as definitions of ILD progression.

Results

An invitation was sent to 40 experts in SAD who were specialists in internal medicine or pulmonology. Twenty-five experts responded: 18 from internal medicine and seven from pulmonology. Forty-eight percent were between 40 and 50 years of age and 20% were over 50 years of age. Seventy-two percent belonged to the Spanish Society of Internal Medicine (SEMI, for its initials in Spanish) and the Systemic Autoimmune Diseases Working Group (GEAS, for its initials in Spanish). Eighty-four percent worked in the public healthcare system and 12% in both the public and private healthcare systems. Seventy percent had been practicing for more than ten years as specialists treating patients with SAD. Ninety-two percent saw more than 20 patients with SAD per month.

General aspects

The panelists considered that screening for ILD should include, at minimum, a case history and pulmonary auscultation, a chest x-ray, and comprehensive respiratory function tests (RFT) (spirometry, lung volumes, and DLCO). Sixty-eight percent also considered high-resolution computed tomography (HRCT) necessary.

For monitoring ILD-SAD progression, the panelists considered performing RFT and repeating the HRCT in case of clinical worsening without abnormalities on the RFT or abnormalities in the RFT without clinical worsening and 6MWT. Table 1 shows the different risk factors for ILD in each of the diseases.

To define the progression of ILD in SAD and assess the transition to the second line of treatment, the panelists were mostly in favor of the definition of ''progressive pulmonary fibrosis'' found in the ATS/ERS/JRS/ALAT guidelines.[2] Finally, it was considered that, in general, follow-up on patients with a diagnosis of ILD-SAD with or without treatment should be performed every three to six months during the first year and then less frequently if the ILD is stable. Table 2 summarizes the method used to define ILD-SAD progression.

The scientific committee's recommendations regarding general aspects of the management of ILD-SAD are summarized in Fig. 1. Table 3 summarizes the general issues.

Interstitial lung disease-systemic sclerosis

In this study, the panelists recommend screening for ILD for all patients with SS, in line with current guidelines.[1,2] The committee was in agreement and considers that screening is necessary for all patients, including those with prescleroderma or initial SS, whether or not they have symptoms. This screening should include a case history, pulmonary auscultation, RFT with DLCO, and a CT scan.[1] The panelists did not reach a consensus regarding when to repeat testing in patients with SS in whom ILD has been initially ruled out. The scientific committee considers that ILD screening can be performed again every six months or one year depending on the risk of progression (Table 4). In line with the panelists, the scientific committee recommends periodically monitoring patients without initial evidence of ILD with RFT and performing HRCT if symptoms appear or worsening of RFT is noted.

The panelists considered it necessary to start treatment in all patients with clinical ILD or ILD with risk factors for progression (demographic; clinical; or according to HRCT, RFT, or serological markers) or patients at high risk of progression according to the 2008 Wells criteria. In regard to therapeutic management, the scientific committee agreed with the panelists' considerations regarding the initiation of treatment, specifically, that all patients with extensive ILD or with limited ILD but with risk factors for progression, regardless of extent, should be treated regardless of the presence of symptoms.

In this regard, the scientific committee recommends MMF or mycophenolic acid as initial treatment for ILD-SS. Rituximab or tocilizumab can be added sequentially to treatment, followed by nintedanib if progression criteria are met. If the patient progresses despite treatment, lung transplantation or, in select patients, hematopoietic stem cell transplantation should be considered. Intravenous cyclophosphamide can be used in case of intolerance to MMF or RTX or TCZ or in cases of rapidly progressive ILD.

Interstitial lung disease-Sjögren's disease

The scientific committee suggests screening for ILD-SjS with RFT in all patients regardless of symptoms. The use of lung ultrasound may be considered if it is available, but it is not mandatory. Performing an HRCT will depend on the results of these tests. In addition, HRCT can be considered in patients with risk factors for ILD-SjS, such as the presence of anti-Ro52 antibodies.

Regarding starting treatment, the panelists as well as the scientific committee concurred with the EULAR 2020 recommendations, which are based on starting pharmacological treatment for patients with ILD-SjS with moderate/severe lung involvement according to ESSDAI.[11] The choice of an

Table 1 Risk factors for interstitial lung disease.

	Greater risk of ILD			Lesser risk of ILD
	Demographic factors	Clinical factors	Serological factors	
Systemic sclerosis[3,46–48]	• Male sex • Older age • Ethnicity: Native American and African ancestry	• Diffuse cutaneous systemic sclerosis • New-onset sclerosis • Respiratory symptoms (dry cough, dyspnea on exertion, bibasilar crackles) • Digital ulcers • Gastrointestinal involvement • Increased MRSS (modified Rodnan skin score)	• Anti-topoisomerase-I antibodies (Scl-70) • Anti-U11/U12 antibodies (RNPC-3) • Anti-SSA • Anti Th/To • Elevated CRP/ESR	• Never smoked • Anti-centromere antibodies
Sjögren's Disease[3]	• Male sex • Older age • Older age of onset of SjS • Longer SjS duration	• Raynaud's disease • Oral ulcers • Focus score ≥4 on salivary gland biopsy.	• ANA • Anti-SSA, Anti-Ro-52 • Anti-SSB • ANCA • Elevated CRP/ESR	
Idiopathic inflammatory myopathies[3,25,73]	• Older age at diagnosis • Black race	• Fever • Arthritis/arthralgias • Antisynthetase syndrome • Clinically amyopathic dermatomyositis • Mechanic's hands	• ANA • Antisynthetase • Anti-Jo1 • Anti-PL7/12 • Anti-MDA5 • Anti-SSA, Anti-Ro52 • Elevated CRP/ESR • Low hemoglobin	• Polymyositis • Cancer • NXP2 antagonists • TIF1g antagonists • Mi2 antagonists
ANCA-associated vasculitis[4,5]			• MPO-ANCA • PR3-ANCA (more frequent in Europe)	
Systemic lupus erythematosus [22]	• Older age • Longer duration of SLE	• Raynaud's disease • Sclerodactyly in scleroderma overlap syndrome	• Anti-La • Anti-Scl-70 • Anti-U1RNP • Anti-Sm • Lupus anticoagulant	

A more extensive bibliography for this table can be found in the full document in the supplementary material.
ILD: interstitial lung disease; SLE: systemic lupus erythematosus; SjD: Sjögren's disease.

appropriate pharmacological approach must take several variables into account, including comorbidities and concomitant organ involvement; however, first and foremost, the specific phenotype of pulmonary involvement must be considered (Table 5).

Interstitial lung disease-idiopathic inflammatory myopathies

Regarding screening, most panelists (62.5%) considered it necessary to screen all patients with IIM for ILD, although the predetermined level of consensus of 75% was not reached. The scientific committee considered it necessary to screen with RFT and CT if there are risk factors for ILD and especially in patients with anti-ARS and anti-MDA5 antibodies (Table 6).

The panelists agreed that first-line treatment of ILD-IIM should include a short course of glucocorticoids and MMF. As second line treatment, it was recommended, in order of preference, to add another immunosuppressant/immunomodulator, to add nintedanib, or to substitute another immunosuppressant/immunomodulator. Along the same lines, the scientific committee recommends MMF or tacrolimus together with glucocorticoids as first-line treatment. In progression or refractoriness, rituximab is recommended in cases in which the radiological pattern of lung

Table 2 Definitions of progression in interstitial lung disease.

ATS/ERS/JRS/ALAT clinical practice guidelines[57]	In a patient with ILD of known or unknown etiology other than IPF who presents with radiological evidence of pulmonary fibrosis, progressive pulmonary fibrosis is defined as the occurrence of at least two of the following three criteria in the past year without an alternative explanation[a]: 1 Worsening of respiratory symptoms 2 Physiologic evidence of disease progression (any of the following): a Absolute decrease in FVC > 5% of the predicted FVC at 1 year of follow-up b Absolute decrease in DLCO (corrected for hemoglobin) >10% predicted at 1 year of follow-up 3 Radiologic evidence of disease progression (one or more of the following): a Increased extent or severity of traction bronchiectasis and bronchiolectasis b New ground-glass opacity with traction bronchiectasis c New fine reticular pattern d Increased extent or coarseness of the reticular anomaly e New or enlarged honeycombing Increase in lobar volume loss
INBUILD Study[7]	**Progressive fibrosing ILD**[b]: • ≥10% relative decrease in FVC (% predicted) • or 5% to <10% relative decrease in FVC (% predicted) and worsening of respiratory symptoms or increased extent of fibrosis on HRCT • or worsening of respiratory symptoms and increased extent of fibrosis on HRCT, regardless of FVC progression (% predicted)
Goh et al.[55]	**Optimal definition of ILD-SS progression**: • Relative decrease in FVC compared to baseline of ≥10% • or a relative decrease in FVC of 5%–9% associated with a relative decrease in DLCO of ≥15%

ALAT: Latin American Thoracic Association; ATS: American Thoracic Society; FVC: forced vital capacity; DLCO: diffusing capacity for carbon monoxide; ILD: interstitial lung disease; ERS: European Respiratory Society; SAD: systemic autoimmune diseases; SS: systemic sclerosis; IPF: idiopathic pulmonary fibrosis; JRS: Japanese Respiratory Society; OMERACT: Outcome Measures in Rheumatology; HRCT: high-resolution computed tomography.

[a] Although it is fundamental to rule out alternative explanations for worsening of these characteristics in all patients with suspected progression, this is especially important in patients with worsening respiratory symptoms and/or decrease in DLCO, given the lower specificity of these circumstances for establishing a diagnosis of progressive pulmonary fibrosis compared to FVC and thoracic CT scan.

[b] The INBUILD trial evaluated the use of the antifibrotic nintedanib in ILD-SAD. As an inclusion criterion in the study, a definition of ''progressive fibrosing ILD'' was proposed that had to be met within the previous 24 months.

involvement is more inflammatory and nintedanib in cases in which it is more fibrosing; combination therapy may also be initiated. In recent years, there is increasing evidence on the efficacy of JAK inhibitors, especially tofacitinib, in patients with dermatomyositis with anti-MDA5 antibodies. Rapidly progressive forms require specific treatment in a specialized unit.

Interstitial lung disease - ANCA-associated vasculitis

The panelists were of the opinion that all patients with AAV should be screened. On the other hand, the scientific committee considers that the approach must be individualized.

Regarding treatment, there are no specific randomized clinical trials to guide the treatment of ILD-AAV.[13] In patients with systemic involvement, the panelists and scientific committee recommend following the international consensus guidelines for the treatment of AAV.[13] When involvement is limited to the lung, treatment is based on the radiological pattern (NSIP or OP vs. UIP) and progression (progressive fibrosing ILD).[13]

Both the panelists and the committee support considering the use of antifibrotics in patients with AAV-ILD and criteria of progressive pulmonary fibrosis.[13] Eighty-seven percent were in favor of the possibility of using avacopan to replace glucocorticoid treatment (Table 7).

Interstitial lung disease-systemic lupus erythematosus

There are no precise recommendations regarding screening. The panelists and scientific committee consider screening necessary in cases of respiratory symptoms or signs. In this disease, manifestation in the form of ILD is usually infrequent. Therefore, there is little evidence regarding therapeutic recommendations. The panelists and scientific committee are in line with these recommendations (Table 8).

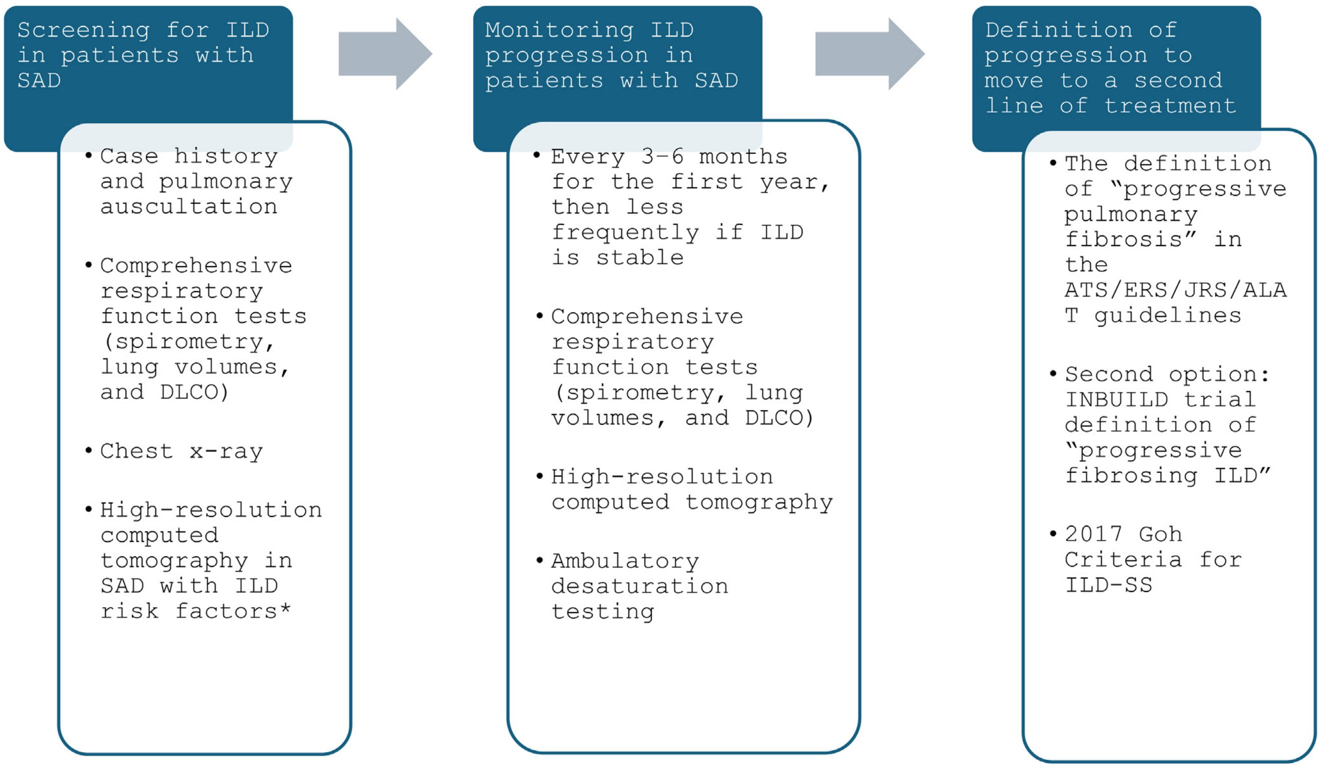

Figure 1 General recommendations of the scientific committee for screening and follow-up of ILD associated with SAD. *Systemic sclerosis, antisynthetase syndrome, ANCA vasculitis, sarcoidosis. ALAT: Latin American Thoracic Association; ATS: American Thoracic Society; DLCO: diffusing capacity for carbon monoxide; ILD: interstitial lung disease; ERS: European Respiratory Society; SAD: systemic autoimmune diseases; JRS: Japanese Respiratory Society.

Sarcoidosis

With respect to the treatments of the forms of ILD in sarcoidosis, the use of antifibrotics associated with immunomodulators in pulmonary fibrotic forms was universally accepted by both the panelists and the committee.

The recommendations of the scientific committee for all forms of ILD-SAD are summarized in Table 9.

Discussion

The aim of this consensus document is to make diagnostic and therapeutic recommendations for the approach to ILD in SAD based on the available evidence and expert opinion. A chest HRCT with a narrow slice width (2 mm) is the most sensitive method for detecting the presence of ILD in any entity.[1] The risk factors for developing ILD depend on each SAD.[3,4] If it has not been performed during screening, it should be done in case of clinical suspicion or abnormalities on the chest X-ray or RFT done during screening.[1,2] It is recommended in case of suspicion of ILD in any SAD, fundamentally in those with SS.[8] However, less than 60% of physicians use HRCT in screening for ILD-SS in patients with newly diagnosed SS.[8] RFT have low sensitivity for detecting ILD-SS and are associated with a high false-negative rate (60% of patients with ILD have baseline RFT > 80% of the predicted value). Therefore, they should not be used as the only screening tests to diagnose ILD-SS and should be used in combination with HRCT.[8] Nonetheless, RFT are very useful during follow-up as the FVC reflects the severity of ILD-SS and is a surrogate marker of mortality.[8] For diagnosis, no other techniques or additional studies are recommended except to rule out associated complications such as infection, suspected alveolar hemorrhage, or neoplasm.[1,2]

In the case of IIM, all patients require a thorough clinical evaluation for possible pulmonary involvement (cough, dyspnea, and decreased exercise tolerance) along with all other extrapulmonary manifestations. Respiratory symptoms may be influenced by concomitant myopathy with involvement of the thoracic muscles of respiration. In all patients with respiratory symptoms, ILD should be ruled out regardless of autoantibody status. In those with anti-ARS and anti-MDA5 antibodies, screening for ILD at disease onset with thoracic HRCT and RFT (with FVC, FEV1, DLCO) and 6MWT is recommended.[12]

With respect to other SAD (SjS, AAV, SLE), the detection of ILD will be based on the presence of symptoms, which should be evaluated at each patient visit.

There are different nomenclatures and criteria for defining the progression of ILD-SAD and the need for antifibrotic treatment.[5–7] The panelists prioritized the one used by the ATS over the INBUILD and Goh criteria (Fig. 1).

Table 3 General issues.

	Favorable opinions		
1.	In general, screening for ILD in patients with SAD should include at minimum	**Anamnesis and pulmonary auscultation**	100%
		Comprehensive respiratory function tests (spirometry, lung volumes, and DLCO)	100%
		Chest x-ray	88%
		High-resolution computed tomography	68%
		Six-minute walk test	48%
		Ambulatory desaturation testing	16%
		Bronchoscopy	4%
		Lung biopsy	4%
2.	Monitoring ILD progression in patients with SAS should generally include	**Comprehensive respiratory function tests (spirometry, lung volumes, and DLCO)**	100%
		High-resolution computed tomography in case of clinical worsening or if its result entails a change of treatment	96%
		Six-minute walk test	80%
		Chest x-ray	48%
		Ambulatory desaturation testing	48%
		High-resolution computed tomography	28%
		Bronchoscopy	4%
		Lung biopsy	0%
3.	In general, the following should be used to define ILD progression in SAD and to assess moving to the second line of treatment	**The definition of ''progressive pulmonary fibrosis'' in the ATS/ERS/JRS/ALAT guidelines**	88%
		The INBUILD trial definition of ''progressive fibrosing ILD''	56%
		OMERACT's definition of ''clinically significant progression''	40%
4.	In general, follow-up on patients with a diagnosis of ILD-SAD with or without treatment should be conducted	**Every 3–6 months for the first year and then more spaced out over time if ILD is stable**	92%
		Every 6–12 months for the first year and more frequently thereafter if ILD is stable	8%

ALAT: Latin American Thoracic Association; ATS: American Thoracic Society; DLCO: diffusing capacity for carbon monoxide; ILD: interstitial lung disease; ERS: European Respiratory Society; SAD: systemic autoimmune diseases; JRS: Japanese Respiratory Society; OMERACT: Outcome Measures in Rheumatology.

Although therapeutic management varies according to the SAD, once the diagnosis of ILD-SAD has been confirmed, it is necessary to assess the severity and evaluate the indication for treatment. Initial treatment usually follows the paradigm of induction and maintenance of remission with immunomodulators. During follow-up, if it meets criteria for progressive pulmonary fibrosis, it may be necessary to add antifibrotic treatment.[1,2]

In the particular case of SS, treatment is considered based on the 2008 Wells progression criteria, which were defined in a cohort of patients with ILD-SS and include HRCT and RFT findings. These criteria consider patients to be at high risk of progression if ILD extent is >20% on HRCT or an indeterminate extent and RFT < 70% (extensive ILD).[9] Regarding first-line treatment of ILD-SS, the 2023 ACR guidelines include MMF as the preferred option, with rituximab and tocilizumab as the main alternatives. As second-line treatment, they recommend adding another immunosuppressant/immunomodulator or switching if there is intolerance, or adding/switching to nintedanib depending on fibrosis on a HRCT.[1] Tocilizumab could be considered in the following situations: 1) second line associated with MMF, especially in early diffuse SS with extrapulmonary manifestations and/or in cases with high acute phase reactants; 2) if MMF is not tolerated; or 3) third line if there is ILD progression despite MMF and nintedanib.[10]

Regarding SjS involvement, glucocorticoid therapy is especially indicated in the more inflammatory ILD patterns (LIP and OP). In NSIP patterns, steroid and immunosuppressive treatment must be administered. Antifibrotic treatments would be more indicated in UIP and progressive fibrosing forms. However, UIP can also be treated with glucocorticoids and immunosuppressants if there is evidence of concomitant inflammatory involvement (such as ground-glass opacity on a CT), especially at diagnosis. In patients with an insufficient response to first-line treatment, the ILD pattern should be reevaluated. In case of predominant inflammation, second-line immunosuppressive treatment with rituximab, calcineurin inhibitors, or cyclophosphamide is advised. Antifibrotic agents are mainly recommended for progressive fibrosing ILD.[11] The 2023 ACR guidelines recommend MMF, azathioprine, or rituximab as first-line treatment in order of preference, although it depends on patient characteristics.[1]

In forms of ILD associated with IIM, treatment is tailored according to the severity of ILD: mild-moderate

Table 4 Interstitial lung disease associated with systemic sclerosis.

	Favorable opinions		
5.	Screening for ILD is recommended	**For all patients with SS**	79%
		Patients with SS and one or more risk factors for developing ILD[a]	17%
		Patients with respiratory symptoms or signs	4%
6.	In patients with SS in whom ILD has initially been ruled out, it is recommended that ILD screening be performed again	Every 6 months or one year depending on risk of progression	58%
		Annually	21%
		Every 6 months for the first 3–5 years and annually thereafter	21%
7.	In patients with SS in whom ILD has initially been ruled out, periodic monitoring is recommended	**With respiratory function tests**	96%
		With high-resolution computed tomography if there is worsening of symptoms or clinically significant progression	96%
		With high-resolution computed tomography in all cases	4%
8.	In patients diagnosed with ILD-SS, starting treatment should be considered in	**Patients at high risk of progression according to the 2008 Wells criteria**[a]	96%
		All patients with clinical ILD	87.5%
		All patients with subclinical ILD and risk factors for progression[b]	87.5%
		All patients	17%
9.	In the first-line treatment of ILD-SS, use of the following may be considered	**Mycophenolate or mycophenolic acid**	96%
		Immunomodulator + nintedanib in cases with rapidly progressive ILD	75%
		Cyclophosphamide	29%
		Tocilizumab	25%
		Nintedanib	21%
		Rituximab	17%
		Azathioprine	4%
10.	The use of glucocorticoids in the initial treatment of ILD-SS is not recommended		75%
11.	Tocilizumab may be recommended as initial treatment for ILD-SS	If there are extrapulmonary manifestations	71%
		If there are high acute-phase reactant levels	71%
		In case of early diffuse SS	62.5%
12.	In patients with ILS-SS who are on adequate treatment with an immunosuppressant/immunomodulator and progression of ILD is observed, the following may be considered	**Adding nintedanib**	100%
		Adding another immunosuppressant/immunomodulator	96%
		Considering hematopoietic stem cell transplantation	79%
		Substituting with another immunosuppressant/immunomodulator	58%
		Substituting with nintedanib	12.5%

ILD: interstitial lung disease; SS: systemic sclerosis.
[a] High risk of progression according to Wells 2008 criteria: extent >20% on high-resolution computed tomography or indeterminate extent and FVC < 70%.
[b] Patients with subclinical ILD and risk factors for progression (demographic, clinical, high-resolution computed tomography, respiratory function tests, or serological markers).

(usually a subacute form), progressive and/or severe (may be subacute or acute), or rapidly progressive, potentially life-threatening.[12] As discussed in the results section, the panelists established a priority when choosing treatment.

Treatment of AAV with generalized systemic involvement, including pulmonary involvement, consists of remission induction with a combination of glucocorticoids and rituximab or cyclophosphamide followed by maintenance therapy with low-dose glucocorticoids and rituximab every four to six months as a first choice or azathioprine, as an alternative, for a minimum of 24 months.[13] Some studies suggest that prolonging the maintenance regimen with immunosuppressants/immunomodulators beyond the general recommendations could prevent ILD progression, although the evidence is not strong.[13] The panelists and scientific committee consider that this strategy can be evaluated in some cases. On the other hand, the presence of MPO plays a role in the pathogenesis of pulmonary fibrosis.[13] Therefore, it seems reasonable to add antifibrotic treatment

Table 5 Interstitial lung disease associated with Sjögren's disease.

	Favorable opinions		
13.	Screening for ILD is recommended	**For patients with SjS with respiratory symptoms or signs**	62.5%
		For all SjS patients	37.5%
14.	Initial screening of all patients with SjS (even if they are asymptomatic) with respiratory function tests and lung ultrasound can be considered		87.5%
15.	Immunosuppressive/immunomodulatory treatment should be considered	**In patients with moderate/severe pulmonary involvement according to the ESSDAI**	100%
		In all patients with ILD-SjS	23%
16.	In the first-line treatment of ILD-SjS, the use of the following can be considered	**Mycophenolate or mycophenolic acid**	96%
		Glucocorticoids	87.5%
		Azathioprine	37.5%
		Rituximab	33%
		Cyclophosphamide	33%
17.	In patients with ILD-SjS who are on adequate treatment with an immunosuppressant/immunomodulator and progression of ILD is observed, the following may be considered:	**Adding another immunosuppressant/immunomodulator**	96%
		Adding nintedanib	92%
		Substituting with another immunosuppressant/immunomodulator	83%
		Substituting with nintedanib	12.5%

ILD: interstitial lung disease; SAD: systemic autoimmune diseases; SjD: Sjögren's disease.

Table 6 Interstitial lung disease associated with idiopathic inflammatory myopathies.

	Favorable opinions		
18.	Screening for ILD is recommended	**For all patients with IIM**	62.5%
		For patients with IIM and demographic, clinical, or laboratory risk factors for developing ILD	33%
		For patients with IIM with respiratory symptoms or signs	4%
19.	In the first-line treatment of ILD-IIM, the use of the following can be considered:	**Glucocorticoids**	92%
		Mycophenolate or mycophenolic acid	83%
		Calcineurin inhibitors	71%
		Azathioprine	42%
		JAK inhibitors in anti-MDA-5 patients	42%
		Cyclophosphamide	37.5%
		Rituximab	33%
20.	In patients with ILD-IIM who are on adequate treatment with an immunosuppressant/immunomodulator and progression of ILD is observed, the following may be considered:	**Adding another immunosuppressant/immunomodulator**	96%
		Adding nintedanib	87.5%
		Substituting with another immunosuppressant/immunomodulator	83%
		Considering hematopoietic stem cell transplantation	42%
		Substituting with nintedanib	12.5%

ILD: interstitial lung disease; IIM: idiopathic inflammatory myopathies.

Table 7 Interstitial lung disease associated with ANCA-associated vasculitis.

	Favorable opinions	
21.	It is recommended that all patients with AAV be screened for ILD	50%
22.	It is recommended to request ANCA in the initial serological tests of all patients with ILD	100%
23.	The treatment of patients with systemic vasculitis is based on the guidelines for treating AAV itself	96%
24.	In the case of ILD-AAV, prolonging the maintenance regimen of immunosuppressants/immunomodulators beyond the time required for remission of vasculitis may be considered to prevent ILD progression	83%
25.	Nintedanib may be considered for use in some cases of ILD-AAV, especially in MPO-ANCA + patients	75%
26.	Avacopan may be considered for replacing glucocorticoid therapy in some cases of ILD-AAV	**87.**50%

ILD: interstitial lung disease; AAV: ANCA-associated vasculitis.

Table 8 Interstitial lung disease associated with systemic lupus erythematosus and sarcoidosis.

	Favorable opinions		
27.	Screening for ILD is recommended	**For patients with SLE with respiratory symptoms or signs**	79%
		For all patients with SLE	21%
28.	In the treatment of ILD-SLE, depending on the severity, the use of the following may be considered:	**Glucocorticoids**	96%
		Mycophenolate or mycophenolic acid	96%
		Cyclophosphamide	79%
		Rituximab	79%
		Azathioprine	42%
		Belimumab	42%
		Intravenous immunoglobulin	42%
		Nintedanib	42%
		Plasmapheresis	
29.	In the treatment of patients with fibrotic pulmonary sarcoidosis, the use of immunosuppressants/immunomodulators associated with antifibrotic drugs may be considered		100%

to immunomodulatory treatment, along the lines discussed by the panelists and the committee.

Evidence on the treatment of ILD in SLE is limited and management recommendations are based on expert opinion.[14] Glucocorticoids and azathioprine/MMF, drugs also used as maintenance therapy, are recommended as first-line treatment for mild and moderate forms. For refractory forms, high doses of glucocorticoids can be used followed by a steroid-sparing agent (e.g. cyclophosphamide). Rituximab or intravenous immunoglobulins have also been used for refractory cases or cases with progressive fibrosis.[14]

In the treatment of patients with sarcoidosis, assessment of pulmonary involvement by function and HRCT is essential. It is usually recommended to initiate glucocorticoids at 0.5 mg/kg/day. In cases of progression or fibrotic pulmonary involvement, the use of oral immunosuppressants, preferably methotrexate or azathioprine in case of pregnancy or intolerance to methotrexate, can be considered. The use of anti-TNF or cyclophosphamide is recommended in severe or accelerated forms.[15]

Conclusions

In this work, a group of experts provides recommendations for the diagnosis and follow-up of patients with ILD-SAD. These recommendations, while not fully addressing the complexity and heterogeneity of these diseases, are intended to serve as a practical tool for clinicians in the management of these complex conditions.

Table 9 Summary of recommendations.

	Systemic sclerosis	Sjögren's disease	Idiopathic inflammatory myopathies	ANCA-associated vasculitis	Systemic lupus erythematosus
ILD screening	• All patients, including patients with presclero-derma or initial SS, whether or not they have symptoms • Screening by means of a case history, pulmonary auscultation, RFT with DLCO and CTAR	• RFT for all patients with SjS with or without respiratory symptoms or signs • Lung ultrasound if available • HRCT depending on initial test results or in high-risk patients (e.g. anti-Ro52+)	• Presence of risk factors for ILD, especially anti-ARS and anti-MDA5 antibodies	• Confirm the diagnosis of ILD in case of suspicion after the usual tests for the study of vasculitis	• Patients with SLE with respiratory symptoms or signs
When to consider starting treatment of ILD	• Patients at high risk of progression according to the 2008 Wells criteria[a] • All patients with clinical ILD • All patients with subclinical ILD and risk factors for progression[b]	• In patients with moder-ate/severe pulmonary involvement according to the ESSDAI	• Generally for all patients • It may be delayed in some cases, when the ILD extension is minimal and non-progressive, asymp-tomatic, and without functional compromise	• The treatment of patients with systemic vasculitis is based on the guidelines for treating AAV itself • Prolonged use of immunosuppres-sants/ immunomodula-tors beyond the time required for remission of vasculitis may be considered to prevent ILD progression	
Preferred first-line treatments	• Mycophenolate or mycophe-nolic acid • The use of glucocorti-coids is not recom-mended	• Glucocorticoids • Mycophenolate or mycophenolic acid	• Glucocorticoids • Mycophenolate/ mycopheno-lic acid or tacrolimus	• Nintedanib may be considered for use in some cases of ILD-AAV, especially in MPO-ANCA + patients	In order of preference: • Glucocorticoids • Mycophenolate or mycophenolic acid • Cyclophosphamide • Rituximab

Table 9 (*Continued*)

	Systemic sclerosis	Sjögren's disease	Idiopathic inflammatory myopathies	ANCA-associated vasculitis	Systemic lupus erythematosus
Management of progression to first-line treatment	• Add rituximab • Add tocilizumab • Adding nintedanib • Considering hematopoietic stem cell transplantation • Consider lung transplantation • Cyclophosphamide if intolerance or rapidly progressing cases	• Add another immunosuppressant/ immunomodulator in more inflammatory patterns or nintedanib in cases of progressive fibrosis	• Add another rituximab or cyclophosphamide in more inflammatory patterns or nintedanib in cases of progressive fibrosis • Treatment of rapidly progressing forms in specialized units		

FVC: forced vital capacity; DLCO: diffusing capacity for carbon monoxide; ILD: interstitial lung disease; SjD: Sjögren's disease; HRCT: high-resolution computed tomography; AAV: ANCA-associated vasculitis.
[a] High risk of progression according to Wells 2008 criteria: extent >20% on high-resolution computed tomography or indeterminate extent and FVC < 70%.
[b] Patients with subclinical ILD and risk factors for progression (demographic, clinical, high-resolution computed tomography, respiratory function tests, or serological markers).

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Appendix A. Supplementary data

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